SET THEORY

Venn Diagrams, Algebra of Sets and Inclusion–Exclusion Principle

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1. VENN DIAGRAM

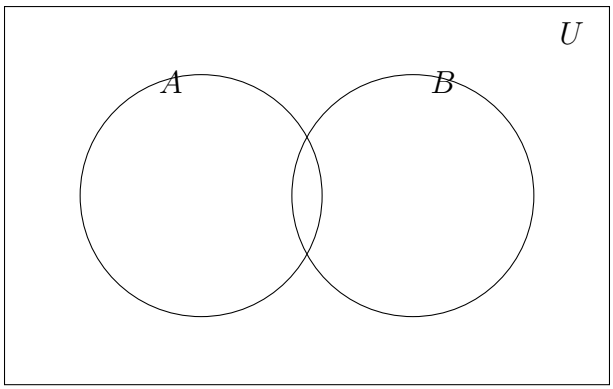
1.1 What is a Venn Diagram?

A **Venn diagram** is a pictorial representation of sets.

It uses:

- a **rectangle** to represent the universal set U,
- **circles or closed curves** to represent sets.

For example, if A and B are two sets, their Venn diagram can be represented as follows:



The overlapping region represents the elements common to both A and B.

Therefore,

$$\boxed{A \cap B}$$

represents the common region.

2. Universal Set

The **universal set** is the set containing all the elements under consideration.

It is usually represented by:

$$\boxed{U}$$

In a Venn diagram, the universal set is represented by a rectangle.

For example, if we are discussing students in a class, then

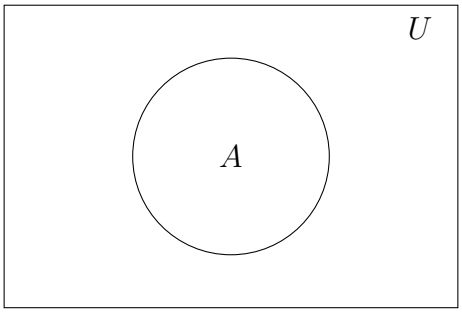
$$U = \{\text{all students in the class}\}.$$

3. Venn Diagram for Basic Set Operations

Students should be able to identify and shade the required regions for the following set operations.

3.1 Set A

To represent A, draw a circle labelled A.



The inside of the circle represents the elements of A.

3.2 Complement of A

The complement of A is written as:

$$\boxed{A'}$$

It means:

All elements in U that are **not in** A.

Therefore,

$$\boxed{A' = U - A}$$

In the Venn diagram, the region outside A is shaded.

3.3 Union of Two Sets

The union of A and B is:

$$\boxed{A \cup B}$$

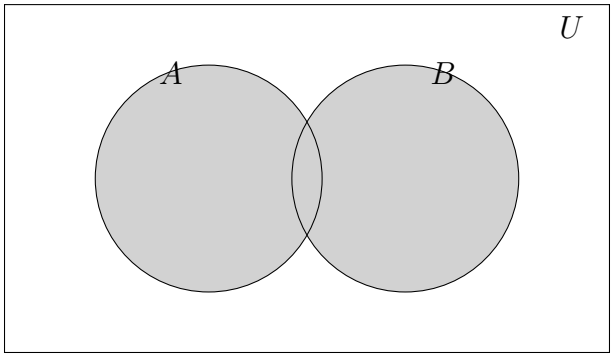
It contains all elements that belong to A, B, or both.
In simple words:

$$\boxed{\cup = \text{OR}}$$

That is,

$$x \in A \cup B$$

if $x \in A$ or $x \in B$.



The complete area of both circles, including the overlapping region, is shaded.

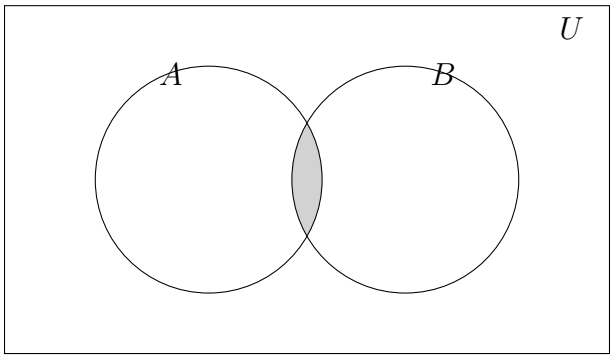
3.4 Intersection of Two Sets

The intersection of A and B is:

$$\boxed{A \cap B}$$

It contains the elements common to both A and B.
In simple words:

$$\boxed{\cap = \text{AND / COMMON}}$$



Only the overlapping region is shaded.

3.5 Difference of Two Sets

The difference of A and B is:

$$\boxed{A - B}$$

It means:

Elements that are in A but not in B.

It can also be written as:

$$\boxed{A - B = A \cap B'}$$

Similarly,

$$\boxed{B - A = B \cap A'}$$

4. Important Venn Diagram Expressions

Expression	Meaning
A	Elements in A
A'	Elements not in A
$A \cup B$	Elements in A or B or both
$A \cap B$	Elements common to A and B
$A - B$	Elements in A but not in B
$B - A$	Elements in B but not in A
$(A \cup B)'$	Elements outside both A and B
$(A \cap B)'$	Elements not common to both

5. How to Draw a Venn Diagram for an Expression

Use the following steps:

1. Identify the universal set U.
2. Draw a rectangle for U.
3. Draw the required circles for the sets.
4. Identify the set operation.
5. Shade only the required region.

Remember:

$$\boxed{\cup \rightarrow \text{OR}}$$

$$\boxed{\cap \rightarrow \text{AND / COMMON}}$$

$$\boxed{' \rightarrow \text{NOT / OUTSIDE}}$$

6. Example: Draw $(A \cup B)'$

First consider:

$$A \cup B$$

This means everything inside A or B.
Therefore,

$$(A \cup B)'$$

means everything outside $A \cup B$.
Hence, the region outside both circles is shaded.
This gives:

$$\boxed{(A \cup B)' = A' \cap B'}$$

7. Example: Draw $A' \cap B'$

Consider:

$$A' \cap B'$$

Here:

- A' means outside A,
- B' means outside B,
- $\cap$ means AND.

Therefore, we need the region that is outside A AND outside B.
Hence,

$$\boxed{A' \cap B' = (A \cup B)'}$$

This is one of De Morgan's laws.

8. ALGEBRA OF SETS

8.1 What is Algebra of Sets?

Algebra of sets is the collection of rules or laws used to simplify and manipulate set expressions.
For example,

$$A \cup B = B \cup A$$

This tells us that changing the order of the sets does not change the result.
These laws can be verified using Venn diagrams.

9. Commutative Laws

The word **commutative** means that changing the order does not change the result.

For union:

$$\boxed{A \cup B = B \cup A}$$

For intersection:

$$\boxed{A \cap B = B \cap A}$$

Example

Let

$$A = \{1, 2, 3\}$$

and

$$B = \{3, 4, 5\}.$$

Then,

$$A \cup B = \{1, 2, 3, 4, 5\}$$

and

$$B \cup A = \{1, 2, 3, 4, 5\}.$$

Therefore,

$$\boxed{A \cup B = B \cup A}$$

10. Associative Laws

The associative law says that grouping does not affect the result.

For union:

$$\boxed{(A \cup B) \cup C = A \cup (B \cup C)}$$

For intersection:

$$\boxed{(A \cap B) \cap C = A \cap (B \cap C)}$$

11. Distributive Laws

There are two important distributive laws.

11.1 Law 1

$$\boxed{A \cap (B \cup C) = (A \cap B) \cup (A \cap C)}$$

11.2 Law 2

$$\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$$

These laws are similar to the distributive laws used in ordinary algebra.
For example,

$$a(b + c) = ab + ac.$$

Similarly,

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

12. Identity Laws

The identity laws are:

$$\boxed{A \cup \varnothing = A}$$

and

$$\boxed{A \cap U = A}$$

Union with an empty set does not add anything.
Therefore,

$$A \cup \varnothing = A.$$

Taking the common elements of A and the universal set gives A.
Therefore,

$$A \cap U = A.$$

13. Domination Laws

$$\boxed{A \cup U = U}$$

and

$$\boxed{A \cap \varnothing = \varnothing}$$

Everything in U is obtained when A is joined with U.
Therefore,

$$A \cup U = U.$$

There are no elements common between A and the empty set.

Therefore,

$$A \cap \varnothing = \varnothing.$$

14. Idempotent Laws

The same set repeated does not change the result.

$$\boxed{A \cup A = A}$$

$$\boxed{A \cap A = A}$$

For example,

$$\{1, 2, 3\} \cup \{1, 2, 3\} = \{1, 2, 3\}.$$

15. Complement Laws

For any set A:

$$\boxed{A \cup A' = U}$$

and

$$\boxed{A \cap A' = \varnothing}$$

Every element is either:

- in A, or
- outside A.

Therefore,

$$A \cup A' = U.$$

An element cannot be both in A and outside A.
Therefore,

$$A \cap A' = \varnothing.$$

16. Double Complement Law

Taking the complement twice gives the original set.

$$\boxed{(A')' = A}$$

In simple words:

> The complement of the complement of a set is the original set.

17. Absorption Laws

The absorption laws are:

$$\boxed{A \cup (A \cap B) = A}$$

and

$$\boxed{A \cap (A \cup B) = A}$$

For example,

$$A \cup (A \cap B)$$

Since $A \cap B$ is already a part of A, adding it to A does not add anything new. Therefore,

$$\boxed{A \cup (A \cap B) = A}$$

18. De Morgan's Laws

There are two important De Morgan's laws.

18.1 First De Morgan's Law

$$\boxed{(A \cup B)' = A' \cap B'}$$

In words:

> Complement of union = intersection of complements.

Thus,

$$\boxed{\cup \to \cap}$$

when taking the complement.

18.2 Second De Morgan's Law

$$\boxed{(A \cap B)' = A' \cup B'}$$

In words:

> Complement of intersection = union of complements.

Thus,

$$\boxed{\cap \to \cup}$$

when taking the complement.

19. Easy Way to Remember De Morgan's Laws

When a complement is applied to a bracket:

1. Put the complement on each individual set.
2. Change $\cup$ to $\cap$.
3. Change $\cap$ to $\cup$.

For example,

$$(A \cup B)'$$

becomes

$$A' \cap B'.$$

Similarly,

$$(A \cap B)'$$

becomes

$$A' \cup B'.$$

20. Summary of Algebra of Sets

Law	**Formula**
Commutative	$A \cup B = B \cup A$ $A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Identity	$A \cup \varnothing = A$ $A \cap U = A$
Domination	$A \cup U = U$ $A \cap \varnothing = \varnothing$
Idempotent	$A \cup A = A$ $A \cap A = A$
Complement	$A \cup A' = U$ $A \cap A' = \varnothing$
Double Complement	$(A')' = A$
Absorption	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$
De Morgan	$(A \cup B)' = A' \cap B'$ $(A \cap B)' = A' \cup B'$

21. INCLUSION–EXCLUSION PRINCIPLE

21.1 Why do we need this principle?

Suppose:

- 20 students study Python.
- 15 students study Java.

If we simply add,

$$20 + 15 = 35,$$

we may get a wrong answer.

Why?
Some students may study both Python and Java.
Such students have been counted twice.
The **Inclusion–Exclusion Principle** helps us avoid this double counting.

22. Inclusion–Exclusion for Two Sets

For two sets A and B:

$$\boxed{n(A \cup B) = n(A) + n(B) - n(A \cap B)}$$

where:

- $n(A)$ = number of elements in A,
- $n(B)$ = number of elements in B,
- $n(A \cap B)$ = number of elements common to A and B,
- $n(A \cup B)$ = number of elements in at least one of the two sets.

23. Why Do We Subtract the Intersection?

Suppose:

$$n(A) = 20$$

$$n(B) = 15$$

and

$$n(A \cap B) = 5.$$

If we calculate:

$$20 + 15 = 35,$$

the 5 common elements have been counted twice.
Therefore, subtract them once:

$$35 - 5 = 30.$$

Thus,

$$\boxed{n(A \cup B) = 30}$$

24. Important Two-Set Formula

Always remember:

$$\boxed{n(A \cup B) = n(A) + n(B) - n(A \cap B)}$$

It can also be rearranged to find the intersection:

$$\boxed{n(A \cap B) = n(A) + n(B) - n(A \cup B)}$$

25. Previous-Year Type Question: Two Sets

Question:
If

$$n(A) = 28,$$

$$n(B) = 32,$$

and

$$n(A \cup B) = 50,$$

find

$$n(A \cap B).$$

Solution:
Using,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B),$$

substitute the values:

$$50 = 28 + 32 - n(A \cap B).$$

Therefore,

$$50 = 60 - n(A \cap B).$$

Hence,

$$n(A \cap B) = 60 - 50.$$

Therefore,

$$\boxed{n(A \cap B) = 10}$$

26. Three-Set Inclusion–Exclusion

For three sets A, B and C:

$$\boxed{\begin{aligned} n(A \cup B \cup C) = {} & n(A) + n(B) + n(C) \\ & - n(A \cap B) - n(B \cap C) - n(A \cap C) \\ & + n(A \cap B \cap C) \end{aligned}}$$

27. Why Does the Three-Set Formula Have This Form?

First add the individual sets:

$$n(A) + n(B) + n(C).$$

The elements common to two sets have been counted twice.
Therefore, subtract:

$$n(A \cap B),$$

$$n(B \cap C),$$

and

$$n(A \cap C).$$

But the elements common to all three sets have now been subtracted too many times.
Therefore, add:

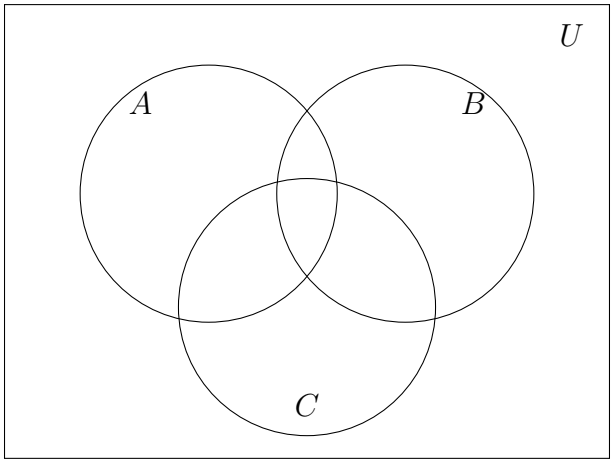
$$n(A \cap B \cap C).$$

Thus, the pattern is:

$$\boxed{+ \text{ individual sets} - \text{ pairwise intersections} + \text{ triple intersection}}$$

28. Three-Set Venn Diagram

For three sets, draw three overlapping circles inside U.



The central region belongs to:

$$\boxed{A \cap B \cap C}$$

29. How to Fill a Three-Set Venn Diagram

A useful method is:

1. Put the value of $A \cap B \cap C$ in the centre.
2. Find the part belonging to exactly two sets.
3. Find the number belonging to only A, only B, and only C.
4. Add all regions to obtain $n(A \cup B \cup C)$.

For example, if

$$n(A \cap B) = 9$$

and

$$n(A \cap B \cap C) = 3,$$

then:

$$\text{Only } A \cap B = 9 - 3 = 6.$$

Similarly,

$$\text{Only } B \cap C = n(B \cap C) - n(A \cap B \cap C)$$

and

$$\text{Only } A \cap C = n(A \cap C) - n(A \cap B \cap C).$$

30. Previous-Year Type Question: Three Sets

Question:
Among a group of students:

- 49 study Physics.
- 37 study English.
- 21 study Biology.
- 9 study Physics and English.
- 5 study English and Biology.
- 4 study Physics and Biology.
- 3 study all three subjects.

Find the number of students in the group.

Solution:
Let

$$P = \text{Physics}, \qquad E = \text{English}, \qquad B = \text{Biology}.$$

Given:

$$n(P) = 49, \quad n(E) = 37, \quad n(B) = 21$$

$$n(P \cap E) = 9, \quad n(E \cap B) = 5, \quad n(P \cap B) = 4$$

$$n(P \cap E \cap B) = 3$$

Using the three-set Inclusion–Exclusion formula:

$$\begin{aligned} n(P \cup E \cup B) = {} & n(P) + n(E) + n(B) \\ & - n(P \cap E) - n(E \cap B) - n(P \cap B) \\ & + n(P \cap E \cap B) \end{aligned}$$

Substituting:

$$= 49 + 37 + 21 - 9 - 5 - 4 + 3$$

$$= 107 - 18 + 3$$

$$= 92$$

Therefore,

$$\boxed{n(P \cup E \cup B) = 92}$$

Hence, the number of students in the group is:

$$\boxed{92}$$

31. Solving the Same Question Using a Venn Diagram

Given:

$$n(P \cap E \cap B) = 3.$$

Put 3 in the centre.

Physics and English only

$$9 - 3 = 6.$$

English and Biology only

$$5 - 3 = 2.$$

Physics and Biology only

$$4 - 3 = 1.$$

Now find only Physics:

$$49 - (6 + 1 + 3)$$

$$= 49 - 10$$

$$\boxed{39}$$

Only English:

$$37 - (6 + 2 + 3)$$

$$= 37 - 11$$

$$\boxed{26}$$

Only Biology:

$$21 - (1 + 2 + 3)$$

$$= 21 - 6$$

$$\boxed{15}$$

Now add all the regions:

$$39 + 26 + 15 + 6 + 2 + 1 + 3$$

$$\boxed{92}$$

Therefore,

$$\boxed{\text{Number of students} = 92}$$

32. Important Terms in Three-Set Problems

32.1 "Study A"

Means:

$$A$$

It may include students who also study other subjects.

32.2 "Study A and B"

Usually means:

$$A \cap B.$$

It may include students who also belong to C.

32.3 "Study all three"

Means:

$$A \cap B \cap C.$$

32.4 "Study at least one"

Means:

$$A \cup B \cup C.$$

32.5 "Study none"

Means:

$$(A \cup B \cup C)'.$$

32.6 "Study only A"

Means students belonging to A, but not B or C.

Symbolically,

$$\boxed{A \cap B' \cap C'}$$

33. Common Mistakes

Mistake 1: Forgetting the overlap

Incorrect:

$$n(A \cup B) = n(A) + n(B)$$

Correct:

$$\boxed{n(A \cup B) = n(A) + n(B) - n(A \cap B)}$$

Mistake 2: Wrong sign for the triple intersection

In the three-set formula, the triple intersection is **added**.

$$\boxed{+n(A \cap B \cap C)}$$

not subtracted.

Mistake 3: Confusing union and intersection

Remember:

$$\boxed{\cup = \text{OR}}$$

$$\boxed{\cap = \text{AND / COMMON}}$$

Mistake 4: Forgetting the universal set

The complement A' is always taken with respect to the universal set U.

Mistake 5: Treating pairwise intersection as "only"

For example,

$$n(A \cap B) = 9$$

does not necessarily mean 9 students belong only to A and B.
Some of those 9 may also belong to C.
If

$$n(A \cap B) = 9$$

and

$$n(A \cap B \cap C) = 3,$$

then:

$$\text{Only } A \cap B = 9 - 3 = 6.$$

34. EXAM-ORIENTED PRACTICE QUESTIONS

34.1 A. Venn Diagram Questions

Draw Venn diagrams for:

a) $A \cup B$

b) $A \cap B$

c) A'

d) $A - B$

e) $(A \cup B)'$

f) $A' \cap B'$

Answer:

$$\boxed{(A \cup B)' = A' \cap B'}$$

Therefore, (e) and (f) have the same shaded region.

34.2 B. Algebra of Sets

Simplify the following.

1.

$$A \cup \varnothing$$

Answer:

$$\boxed{A}$$

2.

$$A \cap U$$

Answer:

$$\boxed{A}$$

3.

$$A \cup A'$$

Answer:

$$\boxed{U}$$

4.

$$A \cap A'$$

Answer:

$$\boxed{\varnothing}$$

5.

$$(A')'$$

Answer:

$$\boxed{A}$$

6.

$$A \cup (A \cap B)$$

Answer:

$$\boxed{A}$$

7.

$$(A \cup B)'$$

Answer:

$$\boxed{A' \cap B'}$$

35. Inclusion–Exclusion Practice

Problem 1

In a class,

$$n(A) = 30, \qquad n(B) = 25$$

and

$$n(A \cap B) = 10.$$

Find $n(A \cup B)$.
Answer:

$$n(A \cup B) = 30 + 25 - 10$$

$$\boxed{45}$$

Problem 2

If

$$n(A) = 40, \qquad n(B) = 35,$$

and

$$n(A \cup B) = 60,$$

find $n(A \cap B)$.
Answer:

$$60 = 40 + 35 - n(A \cap B)$$

$$n(A \cap B) = 75 - 60$$

$$\boxed{15}$$

Problem 3

Given:

$$n(A) = 40, \qquad n(B) = 35, \qquad n(C) = 25,$$

$$n(A \cap B) = 8,$$

$$n(B \cap C) = 6,$$

$$n(A \cap C) = 5,$$

and

$$n(A \cap B \cap C) = 2.$$

Find:

$$n(A \cup B \cup C).$$

Answer:

$$\begin{aligned} n(A \cup B \cup C) &= 40 + 35 + 25 \\ &\quad - 8 - 6 - 5 + 2 \\ &= 100 - 19 + 2 \end{aligned}$$

$$\boxed{83}$$

36. Quick Revision

Venn Diagrams

$$\boxed{\cup = \text{OR}}$$

$$\boxed{\cap = \text{AND / COMMON}}$$

$$\boxed{' = \text{NOT / OUTSIDE}}$$

Important Set Identities

$$\boxed{A \cup \varnothing = A}$$

$$\boxed{A \cap U = A}$$

$$\boxed{A \cup U = U}$$

$$\boxed{A \cap \varnothing = \varnothing}$$

$$\boxed{A \cup A' = U}$$

$$\boxed{A \cap A' = \varnothing}$$

$$\boxed{(A')' = A}$$

De Morgan's Laws

$$\boxed{(A \cup B)' = A' \cap B'}$$

$$\boxed{(A \cap B)' = A' \cup B'}$$

Inclusion–Exclusion for Two Sets

$$\boxed{n(A \cup B) = n(A) + n(B) - n(A \cap B)}$$

Inclusion–Exclusion for Three Sets

$$\boxed{\begin{aligned} n(A \cup B \cup C) = {} & n(A) + n(B) + n(C) \\ & - n(A \cap B) - n(B \cap C) - n(A \cap C) \\ & + n(A \cap B \cap C) \end{aligned}}$$

Remember the pattern:

$$\boxed{\text{Add individual sets} - \text{pairwise intersections} + \text{triple intersection}}$$